COMP3011 Fitness API (1.1.0)

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Fitness tracking API with authentication, exercise search, and workout analytics.

Authentication

Register a new user

REQUEST BODY SCHEMA: application/json

`username` required	string (Username) [3 .. 50] characters
`email` required	string <email> (Email)
`password` required	string (Password) [8 .. 128] characters

Responses

> **201** Successful Response

> **422** Validation Error

POST `/auth/register`

Request samples

Payload

Content type
application/json

Copy

```
{
    "email": "abdullah@example.com",
    "password": "StrongPass123",
    "username": "abdullah"
}
```

Response samples

201 422

Content type
application/json

Copy

```
{
    "email": "abdullah@example.com",
    "id": 1,
    "username": "abdullah"
}
```

Login and receive an access token

REQUEST BODY SCHEMA: application/json

`username` required	string (Username)
`password` required	string (Password)

Responses

> **200** Successful Response

> **422** Validation Error

POST /auth/login

Request samples

Payload

Content type
application/json

Copy

```
{
    "password": "StrongPass123",
    "username": "abdullah"
}
```

Response samples

200 422

Content type
application/json

Copy

```
{
    "access_token": "eyJ1c2VyX2lkIjoxLCJleHAiOjE3MDAwMDAwMDB9.signature",
    "token_type": "bearer"
}
```

Workouts

Create Workout

Create a new workout record for the authenticated user.

AUTHORIZATIONS: > *HTTPBearer*

REQUEST BODY SCHEMA: application/json

`date` required	string <date> (Date)
`workout_type` required	string (Workout Type) [1 .. 100] characters
`duration_min` required	integer (Duration Min) [1 .. 600]
`notes` >	Notes (string) or Notes (null) (Notes)
`exercises` >	Array of objects (Exercises)

Responses

> **201** Successful Response

> **422** Validation Error

POST `/workouts`

Request samples

Payload

Content type
application/json

Copy Expand all Collapse all


```
{
    "date": "2026-03-09",
    "duration_min": 60,
  - "exercises": [
      + { … }
    ],
    "notes": "Chest and shoulders session",
    "workout_type": "Push"
}
```


Response samples

201 422

Content type
application/json

Copy Expand all Collapse all


```
{
    "date": "2026-03-09",
    "duration_min": 60,
  - "exercises": [
      + { … }
    ],
    "id": 1,
    "notes": "Chest and shoulders session",
    "workout_type": "Push"
}
```


List Workouts

Retrieve workout records belonging to the authenticated user.

AUTHORIZATIONS: > *HTTPBearer*

QUERY PARAMETERS

`skip` integer (Skip) >= 0
Default: 0
Number of records to skip

`limit` integer (Limit) [1 .. 200]
Default: 50
Maximum number of records to return

Responses

> **200** Successful Response

> **422** Validation Error

GET /workouts

Response samples

200 422

Content type
application/json

Copy Expand all Collapse all


```
[
 - {
      "date": "2026-03-09",
      "duration_min": 60,
    + "exercises": [ … ],
      "id": 1,
      "notes": "Chest and shoulders session",
      "workout_type": "Push"
   }
```

```
]
```

Suggest a workout plan

Returns a simple workout plan using the exercise dataset based on goal, number of training days, equipment, and difficulty.

QUERY PARAMETERS

`goal`	string (Goal) Default: `"strength"` Training goal: strength, hypertrophy, or general_fitness
`days`	integer (Days) [3 .. 5] Default: `3` Number of training days per week
`equipment` >	Equipment (string) or Equipment (null) (Equipment) Preferred equipment
`difficulty` >	Difficulty (string) or Difficulty (null) (Difficulty) Preferred difficulty level

Responses

> **200** Successful Response

> **422** Validation Error

GET `/workouts/suggest-plan`

Response samples

200 422

Content type
application/json

Copy

```
null
```

Get Workout

Retrieve a specific workout owned by the authenticated user.

AUTHORIZATIONS: > *HTTPBearer*

PATH PARAMETERS

workout_id
required
integer (Workout Id)

Responses

> **200** Successful Response

— **404** Workout not found

> **422** Validation Error

GET /workouts/{workout_id}

Response samples

200 404 422

Content type
application/json

Copy Expand all Collapse all


```
{
    "date": "2026-03-09",
    "duration_min": 60,
  - "exercises": [
      + { … }
    ],
    "id": 1,
    "notes": "Chest and shoulders session",
    "workout_type": "Push"
}
```


Update Workout

Update an existing workout entry owned by the authenticated user.

AUTHORIZATIONS: > *HTTPBearer*

PATH PARAMETERS

workout_id required	integer (Workout Id)

REQUEST BODY SCHEMA: application/json

date >	Date (string) or Date (null) (Date)
workout_type >	Workout Type (string) or Workout Type (null) (Workout Type)
duration_min >	Duration Min (integer) or Duration Min (null) (Duration Min)
notes >	Notes (string) or Notes (null) (Notes)
exercises >	Array of Exercises (objects) or Exercises (null) (Exercises)

Responses

> **200** Successful Response

— **404** Workout not found

> **422** Validation Error

PUT /workouts/{workout_id}

Request samples

Payload

Content type
application/json

Copy Expand all Collapse all

```
{
   "date": "2026-03-10",
   "duration_min": 45,
 - "exercises": [
    + { … }
   ],
   "notes": "Back and biceps session",
   "workout_type": "Pull"
}
```

Response samples

200 404 422

Content type
application/json

Copy Expand all Collapse all

```
{
   "date": "2026-03-09",
```

```
    "duration_min": 60,
  - "exercises": [
      + { … }
    ],
    "id": 1,
    "notes": "Chest and shoulders session",
    "workout_type": "Push"
  }
```


Delete Workout

Delete a workout entry owned by the authenticated user.

AUTHORIZATIONS: > *HTTPBearer*

PATH PARAMETERS

workout_id required	integer (Workout Id)

Responses

— **204** Successful Response

— **404** Workout not found

> **422** Validation Error

DELETE /workouts/{workout_id}

Response samples

404 422

Content type
application/json

Copy

```
{
    "detail": "Workout not found"
}
```

Analytics

Get workout streak statistics

Returns the current streak, longest streak, and total workout days.

Responses

> **200** Successful Response

GET /analytics/streak

Response samples

200

Content type
application/json

Copy

```
null
```

Get weekly workout summary

Returns workout totals and distribution for a given week.

QUERY PARAMETERS

week_start required	string <date> (Week Start) Start date of the week in YYYY-MM-DD format

Responses

> **200** Successful Response

> **422** Validation Error

GET /analytics/weekly-summary

Response samples

200 422

Content type
application/json

Copy

```
null
```

Exercise distribution by body part

Returns the number of exercises available for each body part in the exercise dataset.

Responses

> **200** Successful Response

GET /analytics/exercise-distribution/body-part

Response samples

200

Content type
application/json

Copy Expand all Collapse all

```
[
 - {
      "category": "string",
      "count": 0
   }
]
```

Exercise distribution by equipment

Returns the number of exercises available for each equipment type in the exercise dataset.

Responses

> **200** Successful Response

GET /analytics/exercise-distribution/equipment

Response samples

200

Content type
application/json

Copy Expand all Collapse all

```
[
 - {
      "category": "string",
      "count": 0
   }
]
```

Top rated exercises

Returns the highest-rated exercises from the dataset, with optional filtering.

QUERY PARAMETERS

body_part >	Body Part (string) or Body Part (null) (Body Part) Filter by body part
equipment >	Equipment (string) or Equipment (null) (Equipment) Filter by equipment
difficulty >	Difficulty (string) or Difficulty (null) (Difficulty) Filter by difficulty
exercise_type >	Exercise Type (string) or Exercise Type (null) (Exercise Type)

Filter by exercise type

limit

integer (Limit) [1 .. 50]

Default: 10

Maximum number of exercises to return

Responses

> **200** Successful Response

> **422** Validation Error

GET /analytics/top-rated-exercises

Response samples

200 422

Content type

application/json

Copy Expand all Collapse all


```
[
 - {
      "body_part": "Abdominals",
      "description": "A core exercise targeting the abdominal muscles.",
      "difficulty": "Beginner",
      "equipment": "Body Only",
      "exercise_type": "Strength",
      "id": 1,
      "name": "Crunch",
      "rating": 8.9,
      "rating_desc": "Highly effective for core activation."
   }
]
```

Exercises

Get available exercise filters

Returns available body parts, equipment types, difficulty levels, exercise types, and total exercise count.

Responses

> **200** Successful Response

GET /exercises/meta/filters

Response samples

200

Content type
application/json

Copy Expand all Collapse all


```
{
  - "body_parts": [
        "string"
    ],
  - "equipment": [
        "string"
    ],
```

```
  - "difficulty": [
        "string"
    ],
  - "exercise_types": [
        "string"
    ],
    "total_exercises": 0
}
```

Recommend exercises

Returns recommended exercises based on filters such as body part, equipment, difficulty, and exercise type.

QUERY PARAMETERS

`body_part` >	Body Part (string) or Body Part (null) (Body Part) Filter by body part
`equipment` >	Equipment (string) or Equipment (null) (Equipment) Filter by equipment
`difficulty` >	Difficulty (string) or Difficulty (null) (Difficulty) Filter by difficulty
`exercise_type` >	Exercise Type (string) or Exercise Type (null) (Exercise Type) Filter by exercise type
`limit`	integer (Limit) [1 .. 20] Default: 5 Maximum number of recommended exercises

Responses

> **200** Successful Response

> **422** Validation Error

GET /exercises/recommend

Response samples

200 422

Content type
application/json

Copy Expand all Collapse all


```
[
 - {
      "body_part": "Abdominals",
      "description": "A core exercise targeting the abdominal muscles.",
      "difficulty": "Beginner",
      "equipment": "Body Only",
      "exercise_type": "Strength",
      "id": 1,
      "name": "Crunch",
      "rating": 8.9,
      "rating_desc": "Highly effective for core activation."
   }
]
```


Get exercise by ID

Retrieve a single exercise record by its unique identifier.

PATH PARAMETERS

exercise_id required	integer (Exercise Id)

Responses

> **200** Successful Response

> **422** Validation Error

GET /exercises/{exercise_id}

Response samples

200 422

Content type
application/json

Copy


```
{
    "body_part": "Abdominals",
    "description": "A core exercise targeting the abdominal muscles.",
    "difficulty": "Beginner",
    "equipment": "Body Only",
    "exercise_type": "Strength",
    "id": 1,
    "name": "Crunch",
    "rating": 8.9,
    "rating_desc": "Highly effective for core activation."
}
```


List exercises

Search and filter exercises from the gym dataset.

QUERY PARAMETERS

`q` >	Q (string) or Q (null) (Q) Search by exercise name or description
`body_part` >	Body Part (string) or Body Part (null) (Body Part) Filter by body part
`equipment` >	Equipment (string) or Equipment (null) (Equipment) Filter by equipment
`difficulty` >	Difficulty (string) or Difficulty (null) (Difficulty) Filter by difficulty level
`exercise_type` >	Exercise Type (string) or Exercise Type (null) (Exercise Type) Filter by exercise type
`sort_by` >	Sort By (string) or Sort By (null) (Sort By) Default: `"recommended"` Sort results: recommended, rating_low_high, rating_high_low
`skip`	integer (Skip) `>= 0` Default: `0` Number of records to skip
`limit`	integer (Limit) `[ 1 .. 200 ]` Default: `50` Maximum number of exercises to return

Responses

> **200** Successful Response

> **422** Validation Error

GET /exercises

Response samples

200 422

Content type
application/json

Copy Expand all Collapse all


```
[
 - {
       "body_part": "Abdominals",
       "description": "A core exercise targeting the abdominal muscles.",
       "difficulty": "Beginner",
       "equipment": "Body Only",
       "exercise_type": "Strength",
       "id": 1,
       "name": "Crunch",
       "rating": 8.9,
       "rating_desc": "Highly effective for core activation."
   }
]
```


UI

Dashboard

Responses

> **200** Successful Response

GET /

Response samples

200

Content type
application/json

Copy

```
null
```

System

Health check

Verify that the API is running.

Responses

> **200** Successful Response

GET /health

Response samples

200

Content type
application/json

Copy

```
null
```